

CDC / 2-FF sync

When a signal crosses from one clock domain to another

Safe, unsafe, multi-bit

- Starter: two-FF mode with `async_in` toggling and `clk_dst` stepping
- After a near-edge sample, `q1` may show M for metastable
- One-FF mode is unsafe, `sync_out` comes straight from `q1`
- Multi-bit misuse warns you
- Never fan `q1` into combinational logic in the `dst` domain

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

CDC / 2-FF synchronizer

Async bit into another clock — metastability risk on the first flop, settled value on the second. Concept model, not SPICE.

Starter example: 2-FF sync — toggle `async_in`, step `clk_dst`, watch `q1` (risk) then `q2` (settled). Compare unsafe 1-FF and multi-bit misuse.

Load starter example

Challenges (1/22)

Quiz: CDC. Clock-domain crossing acronym? Answer: CDC

Your answer

Show hintCheckNextIdle

quiz-cdcquiz-metaquiz-2ffquiz-q2startertogglesteprisk-armmeta-steppreset-2ffpreset-1ffpreset-multidemoexplainmode-1ffquiz-buscode-2ffcode-1ffwarn-1ffpropagatehistfull-path

Core ideas

CDC

Signal crosses unrelated clocks — timing not guaranteed.

Metastability

First dst flop can hang between 0/1 after a bad sample.

2-FF

Second flop samples after settle time — use `q2`, not `q1`.

CDC sync starter

learn_digital — cdc sync

Workbook practice

- In the workbook track, draw `async_in`, `q1`, `q2`, and `sync_out` across two `dst` clock edges
- Write the `always` block for a two-FF synchronizer
- Explain why `q2` is the safe output
- Sketch what goes wrong with one flop or with an eight-bit bus through two flops
- Name one pitfall: using `q1` before it has settled

Pitfalls to watch

- Do not treat CDC as a timing exception you can ignore
- Two flops improve MTBF but do not make multi-bit buses safe
- And remember: the browser lab is literacy
- Real designs still need CDC constraints, synchronizer cells

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one two-FF settle wave and one multi-bit caution
- When you are ready, take the short quiz, then continue to the FSM designer lab